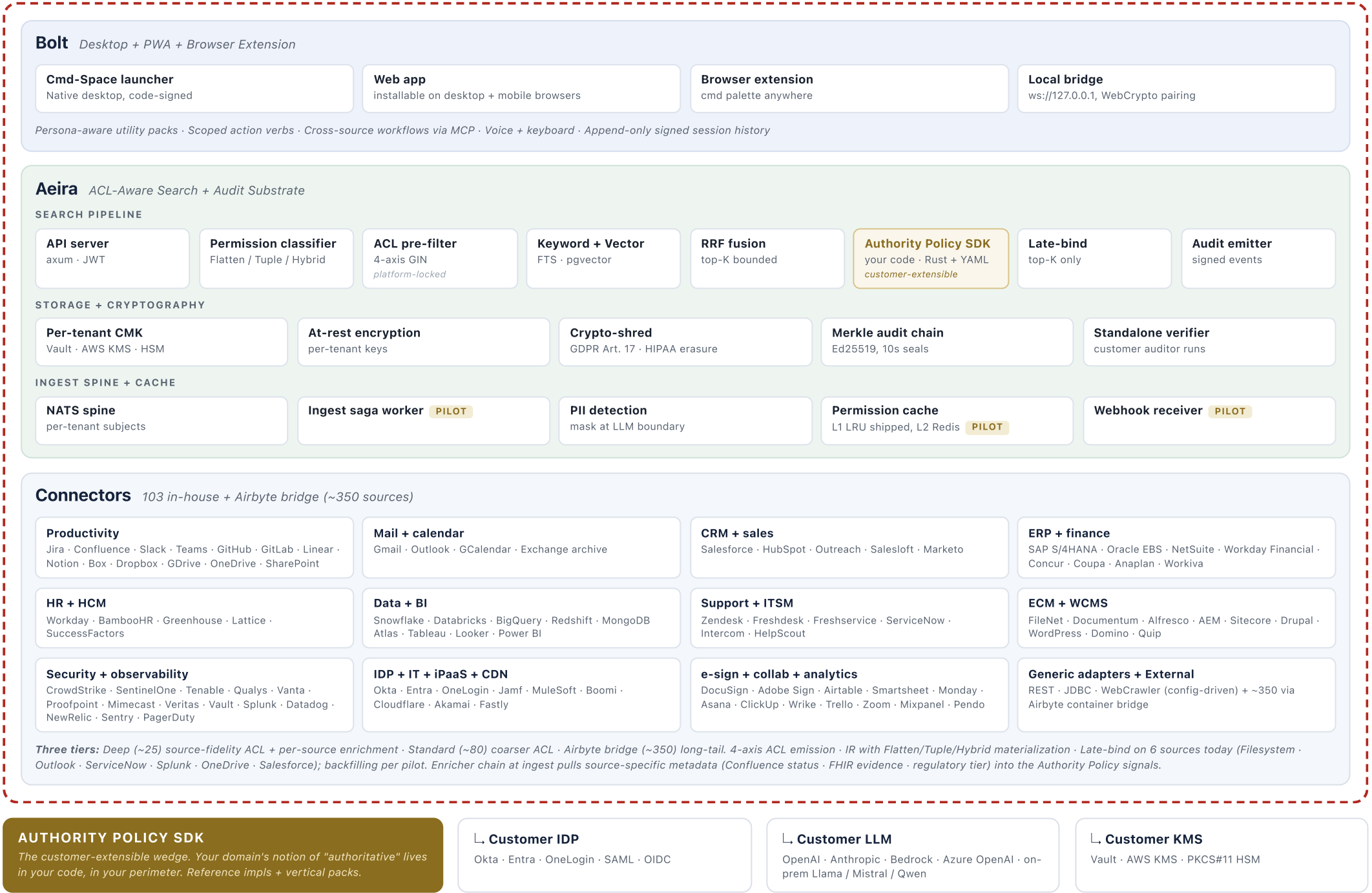


# Bolt + Aeira · Enterprise AI Substrate Architecture

Inside your perimeter. With your LLM. On your audit trail.

— Customer network boundary · All Bolt tiers self-hosted · No outbound to Sparcle · No telemetry —

- 1 **BYO LLM here** — customer's existing endpoint, any vendor.
- 2 **Audit chain seals here** — 10-second Merkle epochs, KMS-signed.
- 3 **ACL pre-filter here** — 4-axis GIN index at query time.
- 4 **Webhook receivers here** — push-mode for top-10 sources **PILOT**
- 5 **Cross-source action verbs via MCP** — Bolt invokes live source APIs.
- 6 **Connector poll here** — per-source cadence, 5 min to hourly.
- 7 **Per-region keyrings** — sovereignty enforced at write + read **PILOT**
- 8 **Permission cache** — L1 LRU shipped; L2 Redis **PILOT**
- 9 **Late-bind check** — top-K only, 6 connectors shipped today.



| TECH STACK  |                                  |
|-------------|----------------------------------|
| language    | Rust 1.85+                       |
| workspace   | ~130 crates                      |
| api         | axum · tower                     |
| runtime     | tokio                            |
| search      | Postgres FTS + pgvector          |
| fusion      | RRF (weighted)                   |
| policy      | Authority Policy SDK             |
| spine       | NATS JetStream                   |
| cache L1    | LRU                              |
| cache L2    | Redis [pilot]                    |
| rebac       | OpenFGA [pilot]                  |
| storage     | FS + S3 Object Lock [pilot]      |
| kms         | Vault · AWS · PKCS#11            |
| auth        | OIDC · SAML · JWT                |
| audit       | Ed25519 + Merkle                 |
| telemetry   | OpenTelemetry · Prom             |
| desktop     | Native · macOS · Linux · Windows |
| web         | Svelte 5 · installable           |
| browser ext | Chrome Manifest v3               |
| mobile      | iOS · Android [Q3 2026+]         |

## Bolt tier guide · Pick by buyer type

**Authorize \$30/seat** · customer self-hosted · BYO LLM · full data sovereignty · regulated & on-prem. **Backbone \$60/seat** · customer self-hosted · BYO LLM · AI orchestration over your keys · PII masking + workflows · regulated mid-market (most popular). **Certify \$90/seat** · customer self-hosted · BYO LLM · SOC 2 readiness + signed DPA/BAA · dedicated delivery + native mobile · audit-ready enterprise. All tiers self-host inside your perimeter — no data reaches Sparcle, and there is no managed-hosting tier.

## ✓ What's shipped today

- 103 in-house source connectors + Airbyte bridge (~350 sources)
- 4-axis ACL pre-filter at index time (provable in audit chain)
- Authority Policy SDK + 3 reference impls (Default · Regulated · Healthcare)
- Sealed Merkle audit chain + standalone verifier binary
- 3 deployment SKUs: trial sidecar · docker-compose · Helm
- 4 KMS variants: Vault · AWS KMS · PKCS#11 HSM · local-dev
- Cross-product audit byte-compatible between Bolt + Aeira

## IDP coverage · Authentication-side

**Native:** Okta · Microsoft Entra · OneLogin · Auth0 · Google Workspace · Zoho. **Via generic OIDC + SAML 2.0** (any compliant IdP): Ping Identity · Keycloak · AWS Cognito · AWS IAM Identity Center · ForgeRock · JumpCloud. **Directory connectors** ship for Okta, Entra, OneLogin; Ping & LDAP/AD on pilot-deliverable list.

## ◆ What ships during pilot

- End-to-end ingest saga (1-2 days from green-light)
- Real-time webhook receivers for top-10 sources
- L2 Redis permission cache + webhook invalidation
- OpenFGA wire-up for tuple-mode ACL
- Late-bind on next 14 connectors (per design partner priority)
- First published latency benchmark (target p95 ≤ 200ms on 100M-doc tenant)